Continuous Convergence and Curried Functional Spaces

1. Preliminary definitions

Definition 1.1. Let X be a set. A *filter* on X is a non-empty set $F \subset 2^X$ which is closed under supersets and finite intersections, and does not contain the empty set. The set of all filters on X will be denoted by $\mathcal{F}(X)$.

Definition 1.2. Let X be a set. A family $\mathcal{B} \subset 2^X$ is called a *filter base* if it does not contain the empty set and is closed under finite intersection. The filter $[\mathcal{B}]$ *generated* by $\mathcal{B}$ is defined as the collection of all supersets of sets from $\mathcal{B}$.
If $\mathcal{B} = \{\{x\}\}$ where $x \in X$, the filter $[x] = [\{\{x\}\}]$ is called the *universal filter* of x.
If $f : X \to Y$ is a map and $F \in \mathcal{F}(X)$ is a filter on X, then $\{f(A) \mid A \in F\}$ is a filter base that generates the *image filter* denoted $f(F)$.

Definition 1.3. Let X be a topological space. Then we say that a filter F on X *converges* to a point $x \in X$, and write $F \to x$, if it contains all open neighborhoods of x.

Remark 1.1. The language of filters is expressive enough to encode all topological properties. For example [1]:

- A sequence $\{x_n\}_{n=1}^{\infty}$ converges to a point $x \in X$ iff the filter
$$F = \{A \subset X \mid x_n \in A \text{ for all but finitely many } n\}$$
converges to x. This filter F is called the *derived filter* of $\{x_n\}_{n=1}^{\infty}$.
- A subset $E \subset X$ is closed if for every point $x \in E$ there is a filter $F \in \mathcal{F}(X)$ converging to x such that $A \cap E \neq \varnothing$ for all $A \in F$.
- A topological space X is compact iff every filter F on X is contained in a convergent filter on X.
- A function $f : X \to Y$ between topological spaces is continuous iff it preserves convergent filters, i.e. $F \to x \in X$ implies $f(F) \to f(x) \in Y$.

Definition 1.4. One may try to decouple the notion of convergent filters from topology. Let X be a set with a relation $\to \subset \mathcal{F}(X) \times X$. Then $(X, \to)$ is called a *convergence space* if

- For all $x \in X$, we have $[x] \to x$;
- If $F \to x$ and $G \to x$, then $F \cap G \to x$;
- If $F \to x$ and $F \subset G$, then $G \to x$.

Definition 1.5. A map $f : X \to Y$ between convergence spaces is said to be *continuous* if $f(F) \to f(x) \in Y$ whenever $F \to x \in X$.

Definition 1.6. Let X, Y be two convergence spaces. Then the product $X \times Y$ can be endowed with the following convergence structure: a filter $F \in \mathcal{F}(X \times Y)$ converges to $(x, y) \in X \times Y$ if and only if $p_X(F) \to x \in X$ and $p_Y(F) \to y \in Y$.

Definition 1.7. Let X, Y be convergence spaces. By $\mathcal{C}(X, Y)$ denote the set of all continuous functions from X to Y. Then $\mathcal{C}(X, Y)$ can be endowed with a convergence structure as follows. For a filter $F \in \mathcal{F}(\mathcal{C}(X, Y))$, we say that $F \to f \in \mathcal{C}(X, Y)$ if and only if $F(P) \to f(p)$ whenever $P \to p \in X$. Here, the filter $F(P)$ is based on

$$\{A(B) \mid A \in F, B \in P\} = \{\{a(b) \mid a \in A, b \in B\} \mid A \in F, B \in P\}.$$

The resulting convergence structure on $\mathcal{C}(X, Y)$ is called the structure of *continuous convergence.*

Remark 1.2. Continuous convergence on $\mathcal{C}(X, Y)$ is the coarsest convergence structure that makes the *evaluation mapping*

$$\omega : \mathcal{C}(X, Y) \times X \to Y$$

continuous. Here, the product $\mathcal{C}(X, Y) \times X$ is endowed with the product convergence structure.

Proposition 1.1. (see [2], page 27): *Let X, Y, Z be convergence spaces. Then the map*

$$\gamma : \mathcal{C}(X \times Y, Z) \to \mathcal{C}(X, \mathcal{C}(Y, Z))$$

defined as $\gamma(f)(x)(y) = f(x, y)$, is well-defined and is a homeomorphism.

Proposition 1.2. (see [2], page 34): *Let X, Y be topological spaces such that X is locally compact and Y is regular. Then the convergence space $\mathcal{C}(X, Y)$ is topologized by the compact-open topology. That is, the convergence relation arising from this topology coincides precisely with the convergence structure given in Definition 1.7.*

Proposition 1.3. *Let X be a convergence space and let Y be a Hausdorff convergence space (i.e. every filter in Y converges to at most one point). Then $\mathcal{C}(X, Y)$ is also a Hausdorff convergence space.*

Proposition 1.4. (see [2], page 29): *Let X be a convergence space and Y be a convergence vector space, i.e. a set endowed with both structures such that the operations $+ : Y \times Y \to Y$ and $\cdot : \mathbb{R} \times Y \to Y$ are continuous. Then $\mathcal{C}(X, Y)$ is also a convergence vector space.*

Corollary 1.1. *If X is a topological space and Y is a topological vector space, then $\mathcal{C}(X, Y)$ is a topological vector space.*

2. Curried continuous functions

Proposition 2.1. *Let X, Y be topological spaces such that X is first countable. Then a sequence $\{f_k\}_{k \in \mathbb{N}} \subset \mathcal{C}(X, Y)$ converges to a function $f \in \mathcal{C}(X, Y)$ (in the sense that its derived filter converges) if and only if $f_k(x_k) \xrightarrow[k \to \infty]{} f(x)$ whenever $x_k \xrightarrow[k \to \infty]{} x$ in X.*

Proof: First, let F be the derived filter of $\{f_k\}_{k \in \mathbb{N}}$ and suppose that F converges to a function $f \in \mathcal{C}(X, Y)$. Consider a sequence $\{x_k\}_{k \in \mathbb{N}}$ converging to $x \in X$, with its derived filter P. Now let V be a neighborhood of $f(x)$ in Y. By the definition of continuous convergence, we have $F(P) \to f(x)$, and so $V \in F(P)$. This means that V contains an element of the base of $F(P)$, namely a set

$$B_{n_0, m_0} = \{f_n(x_m) \mid n \geq n_0,\ m \geq m_0\}.$$

Hence, for $k \geq \max(n_0, m_0)$, we have $f_k(x_k) \in V$, as desired.

Conversely, suppose that $f_k(x_k) \xrightarrow[k\to\infty]{} f(x)$ whenever $x_k \xrightarrow[k\to\infty]{} x \in X$. First let us show that for any subsequence $\left\{f_{n_k}\right\}_{k\in\mathbb{N}}$, we have $f_{n_k}(x_k) \xrightarrow[k\to\infty]{} f(x)$.
To this end, let $n \in \mathbb{N}$. We let $z_n = x_{\varphi(n)}$, where $\varphi(n)$ is the minimal number of those $k \in \mathbb{N}$ that satisfy $n \leq n_k$. We clearly see that $n_1 \geq n_2$ implies $\varphi(n_1) \geq \varphi(n_2)$ and that $\varphi(n_k) = k$ for all $k \in \mathbb{N}$. Therefore, the sequence $\{z_n\}_{n\in\mathbb{N}}$ converges to x. Hence we have $f_k(z_k) \xrightarrow[k\to\infty]{} f(x)$, and so

$$f_{n_k}(x_k) = f_{n_k}\left(z_{n_k}\right) \xrightarrow[k\to\infty]{} f(x). \tag{1}$$

Now, to show that $f_k \to f$ in $\mathcal{C}(X,Y)$, we need to show that the derived filter F generated by the sets $C_n = \{f_k \mid k \geq n\}$ converges to f. To this end, let P be a filter in X converging to a point $x \in X$. Note that since X is first countable, the filter P has a countable base $\{A_m\}_{m\in\mathbb{N}}$ of neighborhoods of x. To show that $F(P)$ converges to $f(x)$, assume the contrary. Then some neighborhood V of $f(x)$ does not lie in $F(P)$, which is to say that for all $n, m \in \mathbb{N}$, we have $C_n(A_m) \not\subset V$. That means that for all $n, m \in \mathbb{N}$, there are $k_m \geq m$ and $x_m \in A_m$ such that $f_{k_n}(x_m) \notin V$.
In particular, we have $f_{k_n}(x_n) \notin V$ for all $n \in \mathbb{N}$. At the same time, since the sets A_n generate P, we see that $x_n \xrightarrow[n\to\infty]{} x$, so we must have $f_{k_n}(x_n) \xrightarrow[n\to\infty]{} f(x)$ by (1), a contradiction. ■

Definition 2.1. Let $n \in \mathbb{N}_0 = \{0, 1, 2, ...\}$. The *space of curried continuous functions* $C_n(\mathbb{R})$ is a Hausdorff topological vector space defined recursively as follows:
- If $n = 0$, we set $C_0(\mathbb{R}) = \mathbb{R}$.
- If $C_n(\mathbb{R})$ is defined, we set $C_{n+1}(\mathbb{R}) = \mathcal{C}(\mathbb{R}, C_n(\mathbb{R}))$. Proposition 1.2, Proposition 1.3, and Corollary 1.1 ensure that $C_{n+1}(\mathbb{R})$ is a Hausdorff topological vector space, provided that $C_n(\mathbb{R})$ is.

Proposition 2.2. *Let* $n \in \mathbb{N}_0$. *Then the map*

$$\gamma : \mathcal{C}(\mathbb{R}^n, \mathbb{R}) \to C_n(\mathbb{R})$$

defined by $\gamma(f)(t_1)(t_2)...(t_n) = f(t_1, ..., t_n)$, *is well-defined and is a homeomorphism.*

Proof: If $n = 0$, the statement is trivial. Assume it is proven for $C_n(\mathbb{R})$. By Proposition 1.1, we have

$$\mathcal{C}(\mathbb{R}^{n+1}, \mathbb{R}) = \mathcal{C}(\mathbb{R} \times \mathbb{R}^n, \mathbb{R}) \cong \mathcal{C}(\mathbb{R}, \mathcal{C}(\mathbb{R}^n, \mathbb{R})) \cong \mathcal{C}(\mathbb{R}, C_n(\mathbb{R})) = C_{n+1}(\mathbb{R}),$$

q.e.d. ■

3. Curried differentiable functions

Definition 3.1. Let $n \in \mathbb{N}_0$. The *space of curried differentiable functions* $D_n(\mathbb{R})$ is a Hausdorff topological vector space defined recursively as follows:
- If $n = 0$, we set $D_0(\mathbb{R}) = C_0(\mathbb{R}) = \mathbb{R}$.
- If the space $D_n(\mathbb{R})$ is defined, we define $D_{n+1}(\mathbb{R})$ as the subspace of $\mathcal{C}(\mathbb{R}, D_n(\mathbb{R}))$ consisting of such functions $f : \mathbb{R} \to D_n(\mathbb{R})$, that there is $f' \in C_{n+1}(\mathbb{R})$ which satisfies

$$f(x_0 + h) = f(x_0) + h \cdot f'(x_0) + o(h) \in C_n(\mathbb{R})$$

for all $x_0 \in \mathbb{R}$.

Definition 3.2. Fix $n \in \mathbb{N}_0$. For all $m \in \mathbb{N}$, the *space of curried m times differentiable functions* $D_n^m(\mathbb{R})$ is defined recursively as follows:

- If $m = 1$, we set $D_n^1(\mathbb{R}) = D_n(\mathbb{R})$.
- If $D_n^m(\mathbb{R})$ is defined, we set $D_n^{m+1}(\mathbb{R})$ to be the space of all functions $f \in D_n(\mathbb{R})$ such that $f' \in D_n^m(\mathbb{R})$.

Trivially, for all $n \in \mathbb{N}_0$, we have

$$D_n^1(\mathbb{R}) \supset D_n^2(\mathbb{R}) \supset D_n^3(\mathbb{R}) \supset \dots$$

Proposition 3.1. *Let $n, m \in \mathbb{N}$, and $f \in D_n^m(\mathbb{R})$. Then for all $x \in \mathbb{R}$, we have $f(x) \in D_{n-1}^m(\mathbb{R})$.*

Proof: … ■

Bibliography

[1] N. Bourbaki, *General Topology, Part I*. 1966.

[2] R. Beattie and H.-P. Butzmann, *Convergence Structures and Applications to Functional Analysis*. 2002.